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MILITARY SPECIFICATION

SPECIMENS, COMPARISON, SURFACE ROUGHNESS (FLAT) WITH CARRYING CASE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers requirements for original and replica-type surface roughness comparison specimens with carrying case.

1.2 Classification. The surface roughness comparison specimens covered by this specification shall be of the following types as specified (see 6.2):

Type I. Original-Type Surface Roughness Comparison Specimens with Carrying Case.

Type II. Replica-Type Surface Roughness Comparison Specimens with Carrying Case.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

KK–L–154 — Leather; Cattlehide, Bag and Case.
NN–P–515 — Plywood, Container Grade
QQ–M–38 — Magnesium alloy die-castings.
PPP–B–585 — Boxes; Wood, Wirebound.
PPP–B–591 — Boxes; Fiberboard, Wood-cleated.
PPP–B–601 — Boxes, Wood, Cleated Plywood.
PPP–B–621 — Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636 — Boxes, Fiber.

MILITARY

MIL–P–116 — Preservation, Methods of.
MIL–I–45177 — Instruments, Tracer, Surface Roughness.

STANDARDS

FEDERAL

FED STD No. 66 — Steel: Chemical Composition and Hardenability.

Federal Test Method Std. No. 151 — Metals; Test Methods.

MILITARY

MIL–STD–10 — Surface Roughness, Waviness and Lay.
MIL–STD–129 — Marking for Shipment and Storage.
MIL–STD–130 — Identification Marking of U. S. Military Property.

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FSC 5220

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MIL–STD–171 (ORD) — Systems for Preparation, Painting, and Finishing Metal and Wood Surfaces.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

U. S. DEPARTMENT OF COMMERCE

Commercial Standard CS8, Gage Blanks.
Supplement to Commercial Standard CS8.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification No. A–3 and No. 13.

(Application for copies of the above publication should be addressed to the American Trucking Associations, Inc. Traffic Department, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

U. S. POST OFFICE DEPARTMENT

Parcel Post Regulations.

(Application for copies should be obtained from any first class U. S. Post Office.)

3. REQUIREMENTS

3.1 Type I. Original-type surface roughness comparison specimens.

3.1.1 Original-type surface roughness comparison specimens shall be rectangular blocks of metal having the characteristics of a typical machined surface on one of the two largest plane surfaces of each specimen and shall be the top of the specimen. The remaining surfaces of each specimen shall have a maximum roughness height rating of sixty-three or better. Specimens shall be in sets of twenty with a specimen for each roughness height rating and type of surface as specified in table I.

TABLE I. *Type I specimens*

Roughness height rating (microinches)	Type of surface [1]	Minimum roughness-width-cutoff (in.)	Lay [2]	Acceptable limits (microinches) (AA or CLA) [3]
2	SF	.030	=	1.5 to 3
4	L	.030	=	3.5 to 5
8	L	.030	=	7 to 10
16	GP	.030	=	15 to 19
16	GF	.030	X	15 to 19
32	GP	.030	=	30 to 37
32	GF	.030	X	30 to 37
63	GP	.030	=	60 to 73
63	S	.030	=	60 to 73
63	EM	.030	C	60 to 73
125	GP	.030	=	118 to 144
125	T	.030	=	118 to 144
125	EM	.100	C	118 to 144
125	MP	.300	|	118 to 144
250	GP	.100	=	238 to 288
250	S	.100	=	238 to 288
250	EM	.300	X	238 to 288
500	GP	.100	=	475 to 575
500	T	.100	=	475 to 575
500	EM	.300	X	475 to 575

[1] See table II—abbreviations.
[2] = Parallel to long dimension of specimen.
X Angular in both directions.
| Parallel to short dimension of specimen.
C Circular.
[3] See 4.1.3.

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3.1.1.1 *Size.* Each individual specimen shall have the following dimensions: length between 1.000 and 1.030 inch, width between .625 and .655 inch, and thickness between .187 and .247 inch.

3.1.1.2 *Material.* Unless otherwise specified, the material for the specimens shall be of type No. 302 of Federal Std. No. 66.

3.1.1.2.1 *Hardness.* The specimens shall have a maximum hardness value of Rockwell C35 or equivalent.

3.1.1.3 *Surface defects.* The roughness comparison surface of each specimen shall be free from imperfections such as stains, flaws, and all other defects.

3.1.1.4 *Carrying case.* The specimens shall be contained in a case composed of two sections, hinged together on one side and provided with a clamping or locking device on the opposite side. There shall be two rows of specimens, five specimens in each row in each section. The specimens shall be arranged in numerical sequence according to the roughness height ratings (see 3.1.1). The interior of the case shall be so constructed that the individual specimens are segregated and securely held in place by strips of metallic material which in turn are held in place by use of suitable size machine screws.

3.1.1.4.1 A brass or other type metallic tag approximately 1 inch by 2 inches in size shall be fastened to the outside cover of each case and shall bear the following commensurate size marking:

Surface Roughness Comparison Specimens
Type I. U. S. Govt. Insp.
5220
Serial No.

Other identifying marking shall be as specified in Standard MIL–STD–130.

3.1.1.4.2 The overall dimensions of the case shall be approximately 4 inches wide by 5 inches long by ¾ inch thick.

3.1.1.4.3 Unless otherwise specified the case shall be of die-cast magnesium-alloy material as specified in Specification QQ–M–38.

3.1.1.4.4 *Case Coating.* A protective coating shall be applied to the entire case as a preservative measure. The processing shall be in accordance with the requirements and in particular with System No. 21.10 as specified in Standard MIL–STD–171 (Ord). The top protective coating shall be of a gray color. Cases which are made from material which is inherently noncorrosive or tarnish proof do not require an additional finish or coating.

3.1.1.5 *Specimen identification.* A brass or other type metallic tag of suitable size shall be fastened in the adjacent area to each specimen. The roughness height rating and the type of comparison surface of the specimen shall be permanently and legibly marked with commensurate size figures on the tag. The bottom of each specimen shall also be marked with the identical data appearing on the adjacent tag. Abbreviations shall be in accordance with table II. Any other abbreviations shall be subject to the approval of the procuring agency.

TABLE II. *Abbreviations*

Superfinish	**SF**
Lapped	**L**
Ground with periphery of wheel	**GP**
Ground with flat side of wheel	**GF**
Shaped	**S**
Turned	**T**
Milled with periphery of cutter	**MP**
End Milled	**EM**

3.1.1.6 Subject to the approval of the procuring agency surfaces produced by machining processes other than those specified in table I may be substituted in the sets (see 6.2).

3.2 Type II. Replica type surface roughness comparison specimens.

3.2.1 Type II. Replica type surface roughness comparison specimens shall be manufactured by electrodepositing a formation of material against a stainless steel pattern rendered conducting. The negative impression formed will be used as a pattern for electroforming a metal deposit which, upon separation, becomes the reproduction of the original master. The largest plane area of each specimen (top surface) shall cover a minimum range from 4 to 500 microinches surface roughness and shall be free from imperfections such as, stains, flaws, etc.

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3.2.2 The replica specimen shall be rectangular in form and shall not be less than ¾ inch in length and $\frac{7}{16}$ inch in width. They shall be situated on a rectangular form referred to herein as a scale.

3.2.3 A minimum of 18 replica specimens shall be on a single scale or a minimum of 9 replica specimens shall be on each of a set of two scales. The replica specimens shall be arranged on the scales in numerical sequence in accordance with their roughness height ratings. The single scale with 18 replica specimens shall be 4½ inches to 6 inches in length and approximately 2½ inches in width. Each of the two scales with 9 replica specimens shall be 4½ inches to 6 inches in length and approximately 1¼ inches in width. All scales shall be $\frac{1}{32}$ to $\frac{1}{16}$ inch in thickness. The tops of the replica specimens shall have the surface roughness height ratings and types of surfaces specified in table III.

TABLE III. *Type II specimens*

Roughness height rating (microinches)	Type of surface [1]	Minimum roughness-width-cutoff (in.)	Lay [2]	Acceptable limits (microinches) (AA or CLA) [3]
4	L	.030	=	2.5 to 5
8	L	.030	=	5.5 to 10
16	GP	.030	=	12 to 19
32	GP	.030	=	26 to 37
32	T	.030	=	26 to 37
32	GF	.030	X	26 to 37
63	GF	.030	X	51 to 73
63	S	.030	=	51 to 73
63	EM	.030	C	51 to 73
125	T	.030	=	100 to 144
125	EM	.100	C	100 to 144
125	MP	.300	|	100 to 144
250	GP	.100	=	200 to 288
250	S	.100	=	200 to 288
250	EM	.300	X	200 to 288
500	GP	.100	=	400 to 575
500	T	.100	=	400 to 575
500	EM	.300	X	400 to 575

[1] See table II—abbreviations.
[2] = Parallel to long dimension of specimen.
X Angular in both directions.
| Parallel to short dimension of specimen.
C Circular.
[3] See 4.1.3.

3.2.4 *Material.* Unless otherwise specified the material for the scale(s) shall be nickel.

3.2.4.1 *Hardness.* The scale(s) shall have a maximum hardness value of Rockwell C40 or equivalent.

3.2.5 *Marking.* The roughness height rating and type of simulated comparison surface for each individual replica specimen shall be permanently and legibly marked with commensurate size figures on the scale adjacent to the specimen. The type of surface may be abbreviated in accordance with table II. Any other abbreviation shall be subject to the approval of the procuring agency.

3.2.6 *Identification.* The following commensurate size markings shall appear on the scale(s):

Surface Roughness Comparison Specimens.
Type II. U. S. Govt. Insp.
5220
Serial No.

Other identifying markings shall be as specified in Standard MIL–STD–130.

3.2.7 *Carrying Case.* A suitable size carrying case shall be provided for each set of scales. The

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carrying case shall be a sheath type, open at one end, and completely enveloping the scale(s) except for cutouts normally provided to facilitate starting the scale(s) from the case and shall have two compartments if the replica specimens are in sets of two. Unless otherwise specified the carrying case shall be of type II, Selection A, Condition B, smooth leather of 5/64 to 8/64 inch in thickness as specified in Specification KK–L–154.

3.2.8 *Workmanship.* Workmanship shall be such that the specimens specified herein shall comply with the requirements in section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The supplier shall inspect each set of specimens (one hundred percent inspection) to determine that the specimens are in accordance with the requirements of this specification and applicable documents.

4.1.2 *Roughness height rating.* The roughness height ratings of the individual specimens shall be determined by an instrument conforming to MIL–I–45177 calibrated for the Arithmetical Average (AA) or Centre Line Average (CLA) as described in Standard MIL–STD–10. No individual readings shall exceed the acceptable limits shown in table I or table III whichever is applicable.

4.1.3 *Carrying case for type I specimens.* The carrying case material for type I specimens shall be inspected in accordance with the requirements of Specification QQ–M–38, unless otherwise specified (see 3.1.1.4.3).

4.1.4 *Carrying case for type II specimens.* Inspection and tests shall be in accordance with the procedure contained in Specification KK–L–154.

4.1.5 The material for type I specimens shall be in accordance with requirements specified in Federal Std. No. 66.

4.1.6 *Hardness testing.* Unless otherwise specified, Method 243 of Federal Test Method Std. No. 151 shall be applicable for hardness values and testing procedures.

4.1.6.1 Hardness of type I specimens shall be determined by tests on surfaces of the specimen other than the top surface.

4.1.7 Deviation from the requirements of table I and table III due to critical supply requirements shall be determined by the contracting officer or procuring agency. The specimens procured under such deviation shall not carry a Federal item identification number. The set identification shall be by a serial number assigned by the procuring agency. This is to assure that specimens not in conformance with the required acceptable limits for roughness height ratings do not enter the Federal Supply System.

4.1.8 The preservation, packaging, and packing of the specimens shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A*

5.1.1.1 *Cleaning.* The specimens shall be cleaned in accordance with process C–1 of Specification MIL–P–116.

5.1.1.2 *Drying.* The specimens shall be dried in accordance with any applicable drying procedure specified in Specification MIL–P–116.

5.1.1.3 *Preservative application.* Type I specimens made from noncorrosive material shall not require a preservative. Type I specimens made from corrosive material shall be preserved with type P–2 preservative as specified in Specification MIL–P–116. Type II specimens shall be preserved with type P–1 preservative as specified in Specification MIL–P–116 and when dry the

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scale(s) shall then be inserted into the carrying case.

5.1.1.4 *Unit packaging.* Type I and II specimens shall be packaged by Method IC–3 as specified in Specification MIL–P–116 and herein.

5.1.2 *Level C.* The specimens shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2). Each container shall contain specimens of one type only.

5.2.1 *Level A.* The carrying case containing the specimens shall be packed in a close-fitting wire-bound box, wood-cleated fiberboard box, wood cleated-plywood box, nail wood box, or a fiberboard box conforming to Specification PPP–B–585, class 3, PPP–B–591 (overseas type), PPP–B–601, style A or B using plywood conforming to Specification NN–P–515, type II or III, class 2, PPP–B–621, class 2, and PPP–B–636, class 2. Closure and strapping requirements shall be in accordance with the applicable box specification or appendix thereto. The gross weight for wood or wood-cleated boxes shall be approximately 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specifications.

5.2.2 *Level B.* The carrying case containing the specimens shall be packed in a close-fitting wire-bound wood box, wood-cleated fiberboard box, wood cleated-plywood box, nail wood box, or a fiberboard box conforming to Specification PPP–B–585, class 1, PPP–B–591 (domestic type), PPP–B–601, (domestic type), PPP–B–621, class 1, PPP–B–636, class 1, respectively. Closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall be approximately 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 *Level C.* The carrying case containing the specimens shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification, Motor Freight Regulations, or Parcel Post Regulations, whichever is applicable.

5.3 Marking. Unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The surface roughness comparison specimens covered by this specification are intended for use in evaluating typical machined surfaces by visual or tactual comparison with the specimen.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of specimens (see 1.2).
(c) Material if different from 3.1.1.2, 3.1.1.4.3, 3.2.4, or 3.2.7.
(d) Place of delivery for acceptance inspection.
(e) Machined processes if different from 3.1.1.6.
(f) Level of preservation, packaging, and packing required.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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